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Full Length Article

Optimizing machine learning for enhanced automated ECG analysis in cardiovascular healthcare

Keyi Tang [a], Shuyuan Ma [a], Xiaohui Sun [b], Dongfang Guo [a,*]

[a] Qingdao Fifth People's Hospital, Shandong 266002, China
[b] Qingdao Central Hospital obstetrical department, Shandong 266002, China

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ABSTRACT

The rapid advancement of connected health technology, exemplified by wearable devices like the Apple Watch, has revolutionized healthcare by enhancing the diagnosis, monitoring, and treatment of various conditions, particularly heart-related issues. However, these devices generate vast amounts of ECG data that require interpretation, underscoring the need for reliable automated ECG analysis methods. This study explores the use of machine learning and deep learning algorithms, including Support Vector Classifier (SVC), RandomForest, XGBoost, and LinearSVC, for ECG classification, aiming to improve accuracy and diagnostic capabilities. While traditional methods rely on heuristic features and shallow architectures, this research focuses on leveraging deep learning architectures to automatically extract relevant features from ECG signals. The proposed approach demonstrates promising results in accurately categorizing heartbeats, offering a potential solution to the limitations of current classification methods. By optimizing classification models with metaheuristic algorithms, such as JADE, the study achieves significant performance improvements, highlighting the effectiveness of integrating advanced optimization techniques into ECG analysis processes. Ultimately, the findings underscore the potential of machine learning and deep learning algorithms in advancing automated ECG analysis for improved cardiovascular healthcare.

1. Introduction

Connected health technology, exemplified by wearable devices like the Apple Watch, represents a groundbreaking advancement in healthcare. These devices have the potential to significantly enhance the diagnosis, monitoring, and treatment of various conditions, particularly heart-related issues [1–4]. Remote monitoring devices have transformed healthcare for individuals with periodic heart arrhythmias by providing continuous monitoring of heart activity. While these devices offer significant benefits, they generate large volumes of ECG data that require interpretation by physicians, highlighting the need for reliable methods to automatically interpret ECGs [5].

The electrocardiogram (ECG) is a non-stationary signal commonly used to evaluate heart rate and rhythm. Classification of ECG signals involves extracting relevant features to ensure accuracy. Swarm optimization algorithms are often used in conjunction with classifiers to find optimal classification parameters for distinguishing between different types of heart conditions based on ECG signal characteristics [6].

Classifying ECG signals is critical in medical applications for diagnosing heart conditions. Presently, machine learning solutions are employed to analyze and classify ECG data [7]. One proposed solution is the use of deep learning architectures, which offer several advantages. In these architectures, convolutional neurons serve as feature extractors, capturing intricate patterns in the ECG signals. These features are then processed by fully connected layers (FCN), which make the final decisions regarding the ECG classes. This approach has shown promise in improving the accuracy of ECG classification by enabling the model to automatically learn relevant features from the data, thus reducing the reliance on handcrafted features and potentially enhancing diagnostic capabilities [8].

Deep learning and machine learning algorithms are being increasingly investigated for a wide range of healthcare applications, with particular emphasis on their ability to detect anomalies in ECG signals for critical patient monitoring [9]. While ECGs are reliable tools for monitoring cardiovascular system functionality, recent attention has shifted towards accurate categorization of heartbeats. However, many studies in this area tend to concentrate on classifying conditions using annotated datasets, rather than focusing on learning and applying transferable knowledge across different tasks [10]. This reliance on

* Corresponding author.
E-mail address: tan-g-ky@163.com (D. Guo).

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Nomenclature

Abbreviations

DE	Differential Evolution
ECG	Electrocardiogram
EnhancedAEO	Enhanced Adaptive Epsilon Constraint-Based Optimization
FCN	Fully-Connected Layers
AEO	Adaptive Epsilon Constraint-Based Optimization
SFFS	Sequential Forward Floating Search
EA	Evolutionary Algorithms
PSO	Particle Swarm Optimization
SI	Swarm Intelligence
WWO	Water Wave Optimization
SVM	Support Vector Machine

Greek and Latin Symbols

SVC	Support Vector Classifier
$\varnothing(x)$	the mapping function
ANN	Artificial Neural Networks
W	weight vector
CNN	Convolutional Neural Network
f(x)	final prediction
MI	Myocardial Infarction
K	total number of trees in the ensemble
RBM	Restricted Boltzmann Machine
T	the space of regression trees
DBN	Deep Belief Networks
f	a function
DL	Deep Learning
$\iota(Y_i, \widehat{Y}_i)$	convex identifiable loss function
RNN	Recurrent Neural Network
$\Omega(F_K)$	Complexity function
LSTM	Long Short-Term Memory
q	tree designs
GRU	Gated Recurrent Unit
Acc	Accuracy
OOB	out-of-bag
Pre	Precision
XGBoost	Extreme Gradient Boosting
FN	False Negatives
GBRT	Gradient Boosting Regressor Tree
TN	True Negatives
PCA	Principal Component Analysis
FP	False Positive
JADE	Adaptive Differential Evolution
TP	True Positives

heuristic features and shallow feature learning architectures can result in suboptimal feature selection and lower classification accuracy [11].

1.1. Literature review

In continuation of this section, recent research has focused on classifying ECG signals using machine learning and deep learning methods. These studies involve collaboration with researchers in the field to explore new techniques and algorithms for improving the accuracy and efficiency of ECG classification.

Mar et al. [12] introduced a sequential forward floating search (SFFS) algorithm for classifying ECG arrhythmias based on a linear discriminant criterion. This algorithm examined a comprehensive set of features and assessed the model's robustness using a multilayer perceptron. The approach surpassed the performance of similar studies while meeting the requirements for ambulatory monitoring. Daamouche et al. [13] introduced an innovative method for generating wavelets that effectively represent ECG beats, emphasizing their discrimination capability. This approach utilizes the polyphase representation of the wavelet filter bank and formulates the design problem within a particle swarm optimization (PSO) framework. Experimental findings, conducted on the MIT/BIH arrhythmia database using a state-of-the-art support vector machine classifier, demonstrated the superior performance of the proposed method in terms of classification accuracy and stability.

Houssein et al. [6] proposed a method for classifying ECG heartbeats that integrates water wave optimization (WWO) with support vector machine (SVM) algorithms. Their work underscores the potential of both artificial neural networks (ANN) and SVM as effective tools for ECG classification. Kachuee et al. [10] devised a deep convolutional neural network (CNN) approach for classifying heartbeats, achieving high accuracy in categorizing five types of arrhythmias. Furthermore, they successfully transferred this knowledge to classify myocardial infarction (MI). The method attained impressive accuracy rates of 93.4 % for arrhythmia classification and 95.9 % for MI classification. These results were validated using PhysionNet's MIT-BIH and PTB Diagnostics datasets. Mathews et al. [14] introduced a deep-learning method for classifying single-lead ECG signals. They utilized the Restricted Boltzmann Machine (RBM) and deep belief networks (DBN) to specifically identify ventricular and supraventricular heartbeats. Remarkably, the algorithm achieved high average recognition accuracies for ventricular ectopic beats (93.63 %) and supraventricular ectopic beats (95.57 %) at a low sampling rate of 114 Hz. This study underscores the potential of deep learning approaches in accurate ECG classification and suggests their applicability to a broader range of physiological signal classifications, including arterial blood pressure, nerve conduction, and heart rate variability studies.

Walsh [1] conducted a study on the application of Support Vector Machine Learning for ECG monitoring, emphasizing its benefits. The research illustrated that Support Vector Machines can proficiently classify ECG signals, showcasing its viability as a machine learning approach. This was evidenced by its successful application to the Kaggle ECG Heartbeat Categorization Dataset. Ebrahimi et al. [9] conducted a comprehensive review of recent deep learning (DL) methods for classifying ECG signals. They explored various DL techniques, including CNN, Deep Belief Network (DBN), Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU). Among these, CNN emerged as the most suitable method for feature extraction, as evidenced by its prevalence in 52 % of the 75 studies conducted between 2017 and 2018. The DL methods showcased impressive accuracy in correctly classifying Atrial Fibrillation (100 %), Supraventricular Ectopic Beats (99.8 %), and Ventricular Ectopic Beats (99.7 %).

Weimann and Conrad [5] employed deep CNN to classify ECG recordings, a process that typically demands costly annotated samples. To mitigate this challenge, they adopted transfer learning, initially pretraining the CNNs on the largest publicly available dataset and then fine-tuning them for Atrial Fibrillation classification. This pretraining step led to a notable improvement in CNN performance, achieving up to an 80 % enhancement while reducing the reliance on annotations. Additionally, the researchers explored both supervised and unsupervised pretraining approaches, highlighting their potential to enhance relevance without the need for expensive ECG annotations.

Ongoing research is continually refining ECG classification methods, leveraging advancements in machine learning and deep learning techniques. While ECGs are dependable tools for cardiovascular monitoring, recent efforts have aimed to enhance heartbeat categorization for more

precise diagnostics. However, a common limitation in many studies is their focus on classifying conditions using annotated datasets, overlooking the potential benefits of learning and applying transferable knowledge across various tasks. This reliance on heuristic features and shallow learning architectures can lead to suboptimal feature selection and reduced classification accuracy, highlighting the need for more robust methodologies. To address these challenges, recent studies have developed methodologies aiming to improve classification performance while also reducing computational resources. These advancements are particularly relevant for applications in ambulatory settings, where efficiency and accuracy are paramount. Despite these efforts, the field still faces challenges in achieving optimal classification accuracy and generalizability across diverse datasets. Further research is needed to develop more adaptable and transferable approaches for ECG classification, ensuring more reliable diagnostics and monitoring in clinical settings.

In this research, hybrid machine learning methods incorporating metaheuristic algorithms were utilized to address the limitations of previous approaches. While machine learning methods like SVC, RandomForest, XGBoost, and LinearSVC have been effective in classification tasks, previous studies did not explore the integration of metaheuristic algorithms to enhance their accuracy. To fill this gap, this study proposes the use of metaheuristic algorithms, specifically the JADE algorithm, to optimize learning models and improve their accuracy and efficiency. Additionally, the study introduces Enhanced AEO, LevyJA, JADE, and OriginalJA algorithms as methods for hyper-parameter optimization of learning models, aiming to further enhance the performance of the machine classification models.

The structure of the subsequent sections in this study is outlined as follows:

Section 2: This section is subdivided into multiple subsections, each dedicated to specific aspects of the strategy. These include an in-depth examination of the models utilized, a detailed explanation of the methodology employed, and a comprehensive assessment of performance metrics. **Section 3:** Within this segment, a thorough comparative study is presented. This study delves into both foundational prediction and classification methodologies, providing a nuanced understanding of their respective strengths and weaknesses. **Section 4:** The concluding segment of this research encapsulates the key insights derived from the study's findings. It serves as a culmination of the research endeavor, offering profound insights and implications drawn from the conducted analyses.

2. Methodology

The research employed a systematic methodology, utilizing the PTB Diagnostic ECG Database for the classification task [15]. The dataset included samples of normal heartbeats and those affected by various arrhythmias. Preprocessing of the datasets involved data cleaning, normalization, and the imputation of missing values with the mean. Subsequently, the data was divided into training and testing sets in a proportionate manner of 70 % and 30 %, respectively. Various machine learning models, including SVC, RandomForest, XGBoost, and LinearSVC, were implemented and evaluated to identify the best-performing model. Following the initial model evaluation, the optimization process focused on refining the parameters of the best-performing model to enhance its predictive capabilities. This optimization included the utilization of optimizer algorithms such as Enhanced AEO, LevyJA, JADE, and OriginalJA. In the final stages of the study, a comprehensive evaluation was conducted, considering metrics such as accuracy, precision, recall, and F1-score, aiming to identify the optimal model and optimizer combination for superior performance in detecting cardiovascular disease. Fig. 1 provides a visual representation of the study's processing flowchart, illustrating the sequential steps undertaken in the research methodology.

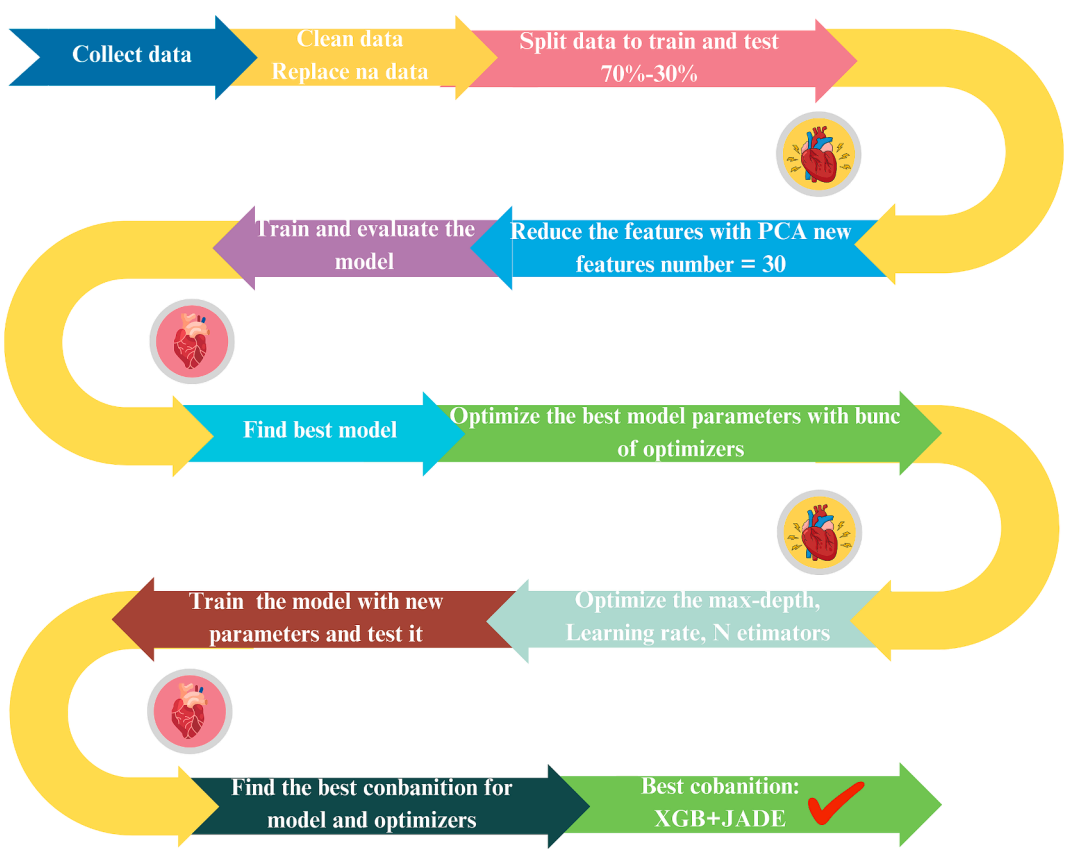

Fig. 1. Depicts the current study's flowchart.

2.1. Machine learning methods

2.1.1. Extreme Gradient Boosting (XGBoost)

Extreme Gradient Boosting (XGBoost) stands at the forefront of machine learning advancements, representing an evolution of the Gradient Boosting Regressor Tree (GBRT) technique. It revolutionizes the landscape by simplifying the complexity of individual regression trees while introducing a highly efficient tree-boosting framework. Renowned for its efficacy, XGBoost has become a staple ensemble technique, amalgamating multiple weak models to forge a formidable predictive model. Central to its innovation is the integration of an objective function with two key components – the loss function and regularization – seamlessly woven into the conventional loss function of GBRT. This integration empowers XGBoost with unparalleled flexibility and adaptability, enabling it to navigate diverse datasets and learning tasks with precision and finesse [16].

XGBoost stands as a powerhouse in supervised learning, tasked with crafting a model that links input parameters to an outcome while minimizing a designated loss function. Its iterative approach involves aligning a weakly trained learner with the gradient of the loss, iteratively honing the final prediction $f(x)$. Through this iterative alignment process, XGBoost progressively enhances the model's predictive prowess, fostering continuous improvement in its ability to make accurate predictions.

$$\widehat{y}_i = \sum_{k=1}^{K} f_k(x_i), f_k \in T \tag{1}$$

here, K denotes the total number of trees in the ensemble. T defines the space of regression trees, with f representing a function within this operational domain T. During model training, XGBoost's target function aims to optimize training effects, striving for improved predictive performance. This objective is encapsulated in the following representation:

$$\widehat{\ell} = \sum_i \iota(Y_i, \widehat{Y}_i) + \sum_K \Omega(F_K) \tag{2}$$

$$\Omega(F_K) = \kappa T + \frac{1}{2}\nu\omega^2 \tag{3}$$

here $\iota(Y_i, \widehat{Y}_i)$ is a convex identifiable loss function that calculates the distinction between predicted actual values for training data. The term $\Omega(F_K)$ defines the complexity of a tree as follows:

$$\ell^{(t)} = \sum_{i=1}^{N} \left[\iota\left(Y_i, \widehat{Y}_i^{(t-1)}\right) + G_i F_t(X_i) + \frac{1}{2} H_i F_t^2(X_i) \right] + \Omega(F_K) \tag{4}$$

which $G_i = \partial_{\widehat{Y}_i^{(t-1)}} \iota\left(Y_i, \widehat{Y}_i^{(t-1)}\right)$ and $H_i = \partial^2_{\widehat{Y}_i^{(t-1)}} \iota\left(Y_i, \widehat{Y}_i^{(t-1)}\right)$.

A greedy strategy is employed to iteratively augment tree designs q by adding branches, commencing with a single leaf. The process entails minimizing loss after each split, which is encapsulated by:

$$\ell_{splite} = \frac{1}{2}\left[\frac{\left(\sum_i \int_{I_L} G_i\right)^2}{\sum_i \int_{I_L} H_i + \nu} + \frac{\left(\sum_i \int_{I_R} G_i\right)^2}{\sum_i \int_{I_R} H_i + \nu} - \frac{\left(\sum_i \int_{I} G_i\right)^2}{\sum_i \int_{I} H_i + \nu} \right] \tag{5}$$

2.1.2. Support Vector Machine

Support Vector Machine (SVM) is a powerful machine learning technique employed in supervised learning tasks to delineate a hyperplane, effectively separating data points. It transcends the confines of linear separability by employing kernel mapping, which elevates feature vectors into higher-dimensional spaces, thereby ensuring the linear separability of samples. At its core, the SVM functions as a linear classifier adept at categorizing data linearly [17]. It surmounts non-linear challenges through the clever utilization of kernel tricks. Its primary objective is to ascertain the optimal hyperplane for both linear and non-linear classification tasks [18].

Widely acknowledged as a robust classification algorithm, the SVM has demonstrated remarkable accuracy in forecasting and addressing regression problems. The training dataset comprises pairs of observations (x_i, y_i), where x_i denotes the input variable and y_i represents the corresponding response for $i = 1, 2, \ldots, m$.

A typical binary of the SVM classifier is formulated as follows:

$$y = f(x) = \omega^T \varnothing(x) + b \tag{6}$$

here $\varnothing(x)$ denotes the mapping function responsible for transforming features from the input space to a higher-dimensional feature space.

The parameter $\frac{\omega^T x + b}{\|W\|}$ calculates the perpendicular distance from the hyperplane to the origin. It represents the Euclidean norm of the weight vector W. Support vectors are determined by the boundary data points used to delineate the margins between the two classes.

The weight coefficient ω and the bias vector b are determined through a process of risk minimization.

$$\min_{w,b} = \frac{1}{2}\omega^2 \tag{7}$$

$$\textit{subjected to } y_i(\ \omega^T\ \varnothing(x_i) + b \geqslant 1, i = 1, 2, \ldots, n \tag{8}$$

In this study, the Support Vector Classifier (SVC) and LinearSVC [1] from the implementation of Support Vector Machines (SVM) are employed for classification tasks. This classifier provides a robust framework capable of handling various types of data, including non-linearly separable datasets, through the utilization of different kernel functions. One notable variant of the SVC is the LinearSVC, which shares similarities with the standard SVC but defaults to using a linear kernel. This characteristic makes LinearSVC particularly well-suited for datasets that exhibit linear separability. Moreover, LinearSVC tends to offer computational advantages over SVC with a linear kernel, especially when dealing with large-scale datasets, as it typically requires less training time.

2.1.3. Random forest

The Random Forest algorithm stands as a cornerstone in the realm of machine learning, renowned for its prowess in both categorization and regression tasks [19]. It operates on the principle of ensemble learning, where a selected model evolves into a conglomerate of diverse models, each offering its unique perspective through individual votes. This collective wisdom is harnessed to make the final decision, yielding superior accuracy compared to standalone components. At its core, the Random Forest method is rooted in the decision tree genre. Here, each tree undergoes a journey of attribute selection at every node, guiding the course of branching decisions. Crafted based on random vectors with uniform distribution across the entire assembly and independently sampled, these trees contribute their distinct classification votes. The ultimate decision is reached through a democratic majority vote, culminating in a robust and reliable prediction [20]. To fine-tune its performance, the Random Forest algorithm relies on pivotal parameters, including the count of variables considered at each tree node and the overall number of trees in the forest. In addition, the concept of out-of-bag (OOB) data serves as an internal arbiter, allowing for validation through error estimation outside the training set. By predicting OOB data through trees, the algorithm unveils a cascade of errors, with the mean out-of-bag error serving as a barometer of the influence of unchosen samples on the final error rate [21].

Practically, the Random Forest algorithm embarks on its journey by selecting a random subset of N data points, constructing decision trees based on this sampled data, and determining the desired number of trees. For regression tasks, each decision tree projects a value as output, which is then aggregated across the entire arboreal assembly through averaging. This meticulous process results in a robust and accurate

regression estimate, showcasing the power and versatility of the Random Forest approach in tackling complex machine learning challenges.

2.2. Principal Component Analysis

Principal Component Analysis (PCA) is a fundamental technique in multivariate analysis, commonly utilized for dimensionality reduction and data exploration [22]. Its primary objective is to transform a dataset of possibly correlated variables into a set of linearly uncorrelated variables called principal components [23]. These components are ordered by the amount of variance they capture, with the first component explaining the maximum variance in the data and each subsequent component explaining the remaining variance in decreasing order. PCA achieves dimensionality reduction by projecting the original data onto a lower-dimensional subspace spanned by the principal components. This transformation not only reduces the dataset's dimensionality but also removes collinearity among variables, simplifies data visualization in two or three dimensions, and helps mitigate the effects of noise in the data. The principal components are computed as linear combinations of the original variables, where the coefficients of these combinations are determined such that the variance of the projected data is maximized. The resulting principal components are orthogonal to each other, meaning they are uncorrelated. This orthogonality property simplifies the interpretation of the transformed data and facilitates further analysis [24].

2.3. Optimization methods

2.3.1. JADE

JADE (Adaptive Differential Evolution with Optional External Archive) stands as a prominent member of the Differential Evolution (DE) family, tailored for addressing continuous optimization challenges. Its primary objective is to determine the optimal set of parameters for a given objective function. Notably, JADE distinguishes itself through its adaptive adjustment of settings throughout the optimization process, a feature that enhances its performance [25].

Renowned for its flexibility and robustness, JADE dynamically adapts its settings to the intricacies of the optimization problem at hand. This adaptive capability often leads to efficient convergence and the generation of high-quality solutions. As a result, JADE has garnered widespread acclaim and serves as a preferred choice for a diverse array of optimization tasks [26].

2.3.2. Enhanced AEO

The EnhancedAEO (Enhanced Adaptive Epsilon Constraint-Based Optimization) [27] emerges as an evolution of the original AEO (Adaptive Epsilon Constraint-Based Optimization) algorithm, aimed at augmenting its performance. While the AEO algorithm unfolds across three distinct phases – production, consumption, and decomposition – the EAEO refines its focus to the production and consumption stages. It is widely acknowledged that achieving an intricate equilibrium between these two conflicting phases significantly boosts the efficacy of metaheuristic algorithms.

The initial phase, termed exploitation, prioritizes local exploration, and refining solutions within a confined vicinity. Conversely, the subsequent phase, exploration, emphasizes global exploration, seeking solutions across a broader landscape. Striking the delicate balance between these phases is paramount, enabling the algorithm to unearth global minima while warding off the entrapment in local optima. This equilibrium fosters a more potent optimization technique, heralding improved performance and efficacy [28].

2.3.3. LevyJA

Jaya, an optimization algorithm proposed by Rao [29], derives its name from the Sanskrit word for "victory." This algorithm stands out for its simplicity, requiring only two parameters: population size and maximum number of generations. Operating on the principle that each solution occupies a specific "position" in the search space, Jaya aims to adjust these positions iteratively to guide solutions toward the optimum. Central to Jaya's approach is the notion that each solution should progressively move away from the worst solution and closer to the best one during the search process. This iterative adjustment process helps refine the solutions over successive generations [30].

In the context of animal behavior, flying animals exhibit Lévy flight behavior, characterized by a random walk pattern with a heavy-tailed probability density function. This behavior involves occasional large steps interspersed with frequent small steps, leading to a motion pattern characterized by local movements, small steps, and occasional large steps. Lévy flights are random walks with stable step distributions, featuring probability distributions with heavy tails. Unlike exponential distributions, which have tails that decay rapidly, the tails of Lévy distributions decay more slowly, allowing for the occurrence of rare but significant events. In the context of multidimensional spaces, such walks exhibit isotropic random patterns, irrespective of the dimensionality of the space [31].

2.3.4. OriginalJA

The Jaya algorithm, introduced by Venkata Rao in 2016, is acclaimed for its stability, rapid convergence, and high accuracy. It surpasses traditional optimization algorithms in numerous engineering challenges, showcasing its effectiveness and superiority. OriginalJA stands out as a robust optimization algorithm tailored for both constrained and unconstrained optimization tasks. It prioritizes superior solutions while actively avoiding inferior ones, all without the need for algorithm-specific parameters [32]. Its simplicity belies its effectiveness, making it a formidable tool in the optimization domain. Unlike Evolutionary Algorithms (EA) and swarm intelligence (SI) algorithms, which often require meticulous tuning of algorithm-specific parameters, OriginalJA eliminates this need. Its straightforward implementation and lack of explicit parameter tuning make it an attractive choice for optimization problems. OriginalJA's simplicity, efficiency, and consistent performance make it the preferred option for a wide range of optimization challenges.

The OriginalJA algorithm has consistently delivered exceptional results, offering global solutions for both continuous and discrete optimization problems with minimal computational burden. Its ease of execution and adaptability to any programming language further enhance its appeal, cementing its status as a premier choice for optimization tasks [33].

2.4. Datasets

The dataset, sourced from the PTB Diagnostic ECG Database, samples including normal heartbeats and those affected by various arrhythmias, as well as signals from different types of myocardial infarctions. These samples have been meticulously curated to provide a comprehensive representation of heartbeat patterns in both healthy and pathological conditions [1]. For researchers and practitioners interested in exploring this dataset further, additional details can be found at the following link: https://www.kaggle.com/datasets/shayanfazeli/heartbeat.

Utilized in research focused on heartbeat classification using machine learning methodologies, this dataset has undergone rigorous preprocessing and segmentation. Each segment of the dataset corresponds to a single heartbeat, allowing for detailed analysis of individual cardiac cycles. The heartbeat type for each sample is denoted in the last column of each row, with beat types represented by integers. The classes include Normal (0), Supraventricular (1), Ventricular (2), Fusion (3), and Unclassified (4), providing a granular classification of heartbeat patterns.

Abnormal heartbeats, such as those associated with supraventricular tachycardia, are characterized by an excessively rapid heart rhythm

resulting from irregular electrical activity in the upper chambers of the heart. The inclusion of such pathological cases in the dataset enhances its utility for studying and understanding cardiac arrhythmias.

The preprocessing steps applied to the dataset:

- To ensure data integrity, any missing values (NaN) within the dataset were identified and imputed using the mean values of their respective columns. For instance, if a particular column that represented heart rate had missing entries, the mean heart rate value across all available samples in that column would be computed and utilized to replace the missing values.
- Prior to model training, the dataset was randomly shuffled to remove any inherent bias in the order of data points. Subsequently, the shuffled dataset was divided into two distinct sets: a training set comprising 70 % of the data and a testing set containing the remaining 30 %.
- To reduce the dimensionality of the dataset and focus on the most informative features, Principal Component Analysis (PCA) was employed. This technique transformed the original dataset into a new set of orthogonal features, with the aim of retaining 30 principal components that capture the maximum variance in the data.
- A heatmap depicting the correlation between different features in the dataset was generated and saved in two formats: as a CSV file (heatmap.csv) and as a PNG image (heatmap.png).
- To maintain a balanced distribution of classes in both the training and testing sets, the number of samples for each class was recorded and saved. This information was stored in separate files for the training (train_classes.csv) and testing (test_classes.csv) sets.

The distribution of samples across different classes in the training and testing sets is visualized in Figs. 2 and 3. The plots reveal that the number of samples in class 0 (normal) is higher compared to other classes, indicating an imbalanced distribution of classes. Fig. 2 illustrates the class distribution in the training set, highlighting the higher number of samples in class 0 (normal) compared to other classes. Similarly, Fig. 3 depicts the class distribution in the testing set.

It must be highlighted here that the resampling of this dataset posed a number of issues. Mainly, there is a high risk of overfitting in models with oversampling in minority classes, since they may learn from duplicated data instead of real patterns. Conversely, any undersampling of the majority class may include the majority of normal heartbeats, or class 0, and thus yield a loss of important diagnostic information and generally degrade the model's performance. In this regard, there was no requirement to make any distribution adjustment in the dataset for the present evaluation.

The heatmap in Fig. 4 represents the individual values of a dataset arranged in a matrix. Heatmaps are valuable tools for visualizing data, especially for identifying patterns or trends across two dimensions. In this heatmap, positive values in the matrix are depicted by colors indicating magnitude, with small numbers around one. As the values increase, they transition to negative numbers. Notably, the values around the diagonal of positive numbers also hover around one, with the width of this range expanding as the numbers increase. This visualization provides a clear overview of the dataset, highlighting variations in values and potential correlations between different elements. It helps in pinpointing areas of interest within the dataset for further analysis and exploration.

2.5. Evaluation of the performance

The primary metric employed to evaluate the performance of classification models in this study is outlined as follows:

2.5.1. Accuracy (Acc)

Accuracy is a fundamental metric that quantifies the ratio of correct predictions to the total number of predictions made by a model. It offers a comprehensive assessment of the model's overall correctness, encompassing predictions across all classes. A higher accuracy score indicates a greater proportion of correct predictions relative to the total predictions made by the model.

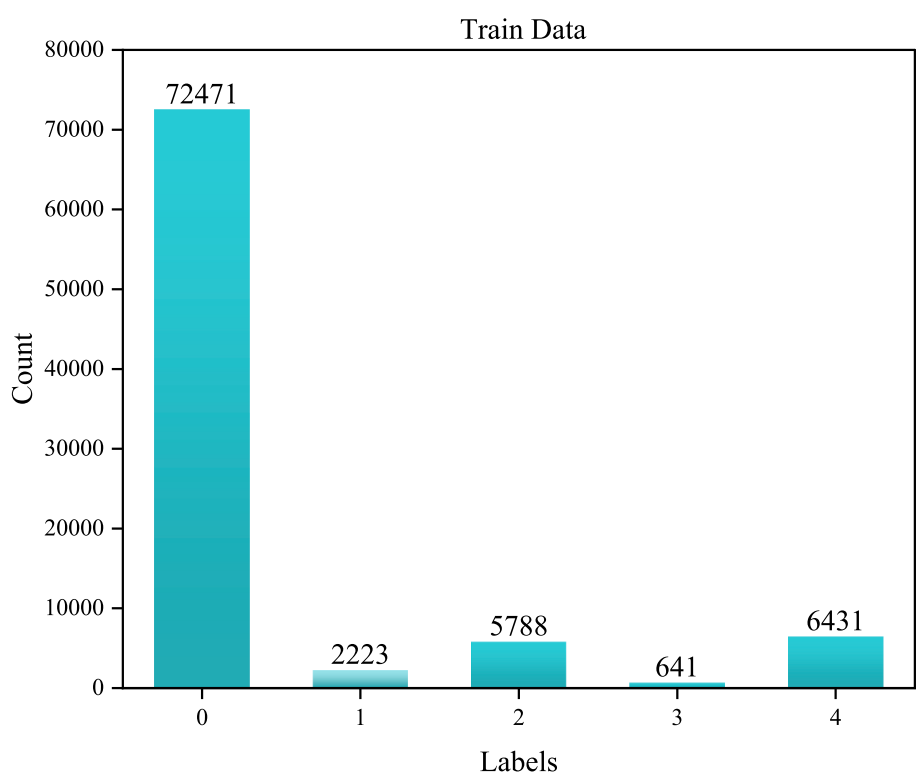

Fig. 2. The number of samples in each training class.

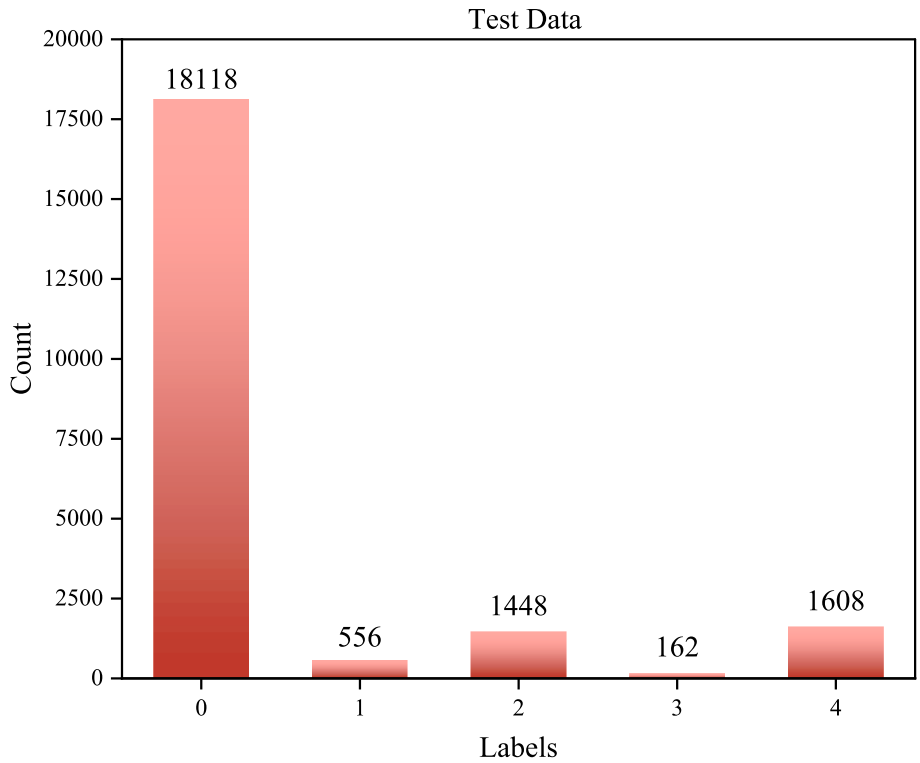

Fig. 3. The number of samples in each testing class.

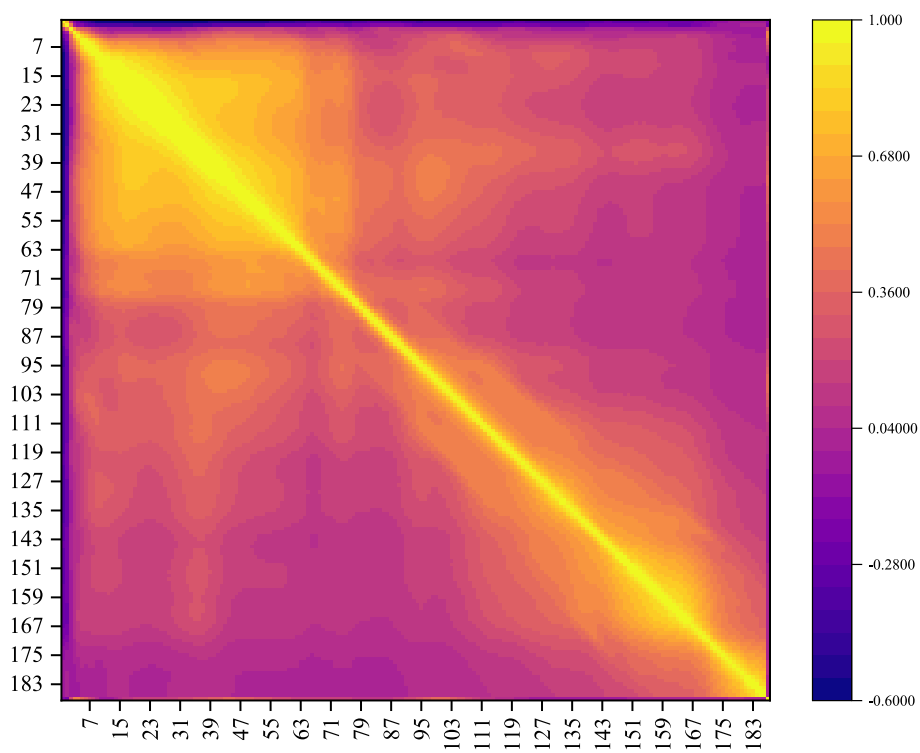

Fig. 4. The heatmap of datasets.

$$\mathrm{Acc} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{9}$$

Here:

TP (True Positive): The count of positive cases correctly identified by the classifier.

TN (True Negative): The number of negative instances accurately predicted by the classifier.

FP (False Positive): The number of actual negative cases incorrectly classified as positive by the classifier.

FN (False Negative): The instances of genuine positive cases wrongly

predicted as negative by the classifier.

2.5.2. Precision (Pre)

Precision, in the context of classification, quantifies the proportion of correctly identified positive cases among all the instances classified as positive by the model. It emphasizes the minimization of false positives, indicating the accuracy of positive predictions made by the classifier.

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{10}$$

2.5.3. Recall

Recall, often referred to as sensitivity or true positive rate, measures the proportion of correctly identified positive cases out of all the actual positive instances in the dataset. It evaluates the classifier's ability to capture and correctly classify all positive instances, emphasizing the minimization of false negatives.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{11}$$

2.5.4. F1-Score

The F1-Score, a harmonic mean of precision and recall, offers a balanced assessment of a classifier's performance, especially in situations with imbalanced class distributions. It considers both precision and recall, providing a single metric to gauge the classifier's overall effectiveness in correctly identifying positive instances while minimizing false positives and false negatives.

$$\text{F1} - \text{Score} = \frac{2 \cdot \text{Pre} \cdot \text{Recall}}{\text{Pre} + \text{Recall}} \tag{12}$$

3. Results and discussion

In this research, the classification of ECG datasets was conducted using four different models: SVC, RandomForest, XGBoost, and LinearSVC. Before applying these models, Principal Component Analysis (PCA) was used to reduce the dataset's dimensionality to 30 features. This step was crucial for managing the data's complexity and improving the models' performance. The evaluation of the model's performance was based on the F1 score rather than accuracy, as the data was unbalanced. The F1 score is a more reliable metric for unbalanced datasets, as it considers both precision and recall. The results indicated that XGBoost achieved the highest F1 score among the four models, demonstrating its superiority in classifying ECG datasets. To further enhance the XGBoost model's performance, metaheuristic algorithms were utilized to optimize its hyperparameters. These algorithms included Enhanced AEO, LevyJA, JADE, and OriginalJA. By optimizing the hyperparameters, the goal was to improve the XGBoost model's ability to accurately classify ECG datasets and potentially outperform its default configuration.

In this section of the study, a detailed evaluation of the results will be presented. This evaluation will provide insights into the effectiveness of the PCA preprocessing step, the performance of the different models, and the impact of hyperparameter optimization on the XGBoost model's classification performance.

Fig. 5 offers a comprehensive overview of the performance metrics obtained from the classification models post-PCA analysis of the datasets. Upon careful examination, it becomes evident that the LinearSVC model displays the least favorable performance compared to the other models. Conversely, both the SVC and RandomForest models exhibit relatively superior performance, showcasing commendable accuracy and efficacy. Notably, the XGBoost model emerges as the top performer among all, showcasing exceptional capabilities in accurately classifying the ECG datasets. This nuanced analysis emphasizes the importance of utilizing the F1-score metric, especially when dealing with imbalanced data, to ensure a comprehensive assessment of the models' performance and robustness.

In the subsequent phase of this study, advanced metaheuristic optimization algorithms, namely Enhanced AEO, LevyJA, JADE, and OriginalJA, were meticulously applied to fine-tune essential hyperparameters such as the learning rate, N-estimators, and max depth. Fig. 6 intricately illustrates the trajectory of training and testing datasets throughout the hybrid classification process employing XGBoost models. Upon close examination of the figures, a consistent trend emerges: while performance metrics across all models remain steady within the training dataset, discernible disparities surface within the testing dataset. Initially, the XGBoost model displays relatively lower

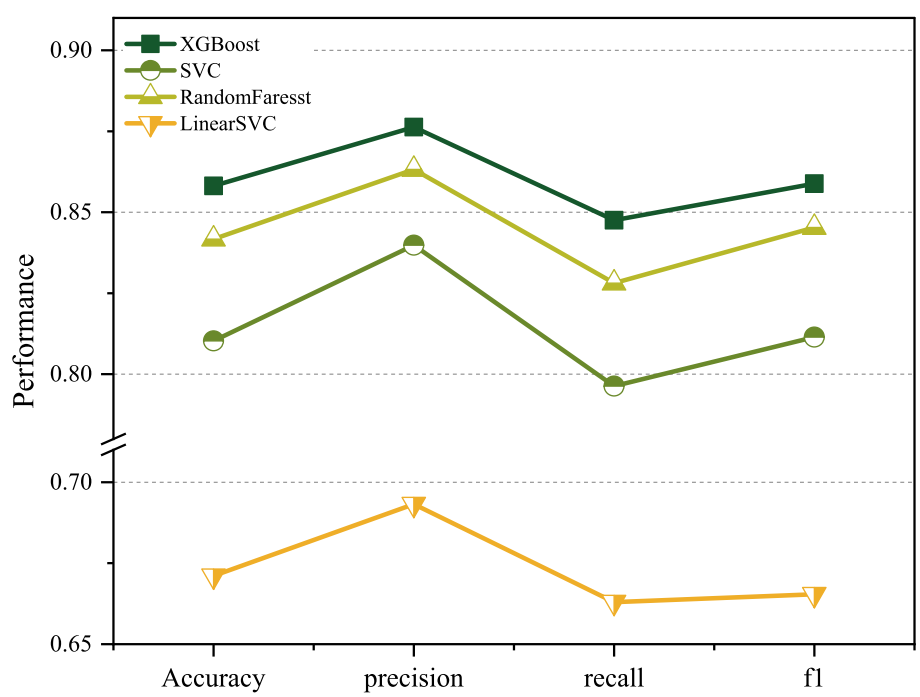

Fig. 5. The performance metrics obtained from XGboost, SVC, random forest, and linearSVC models.

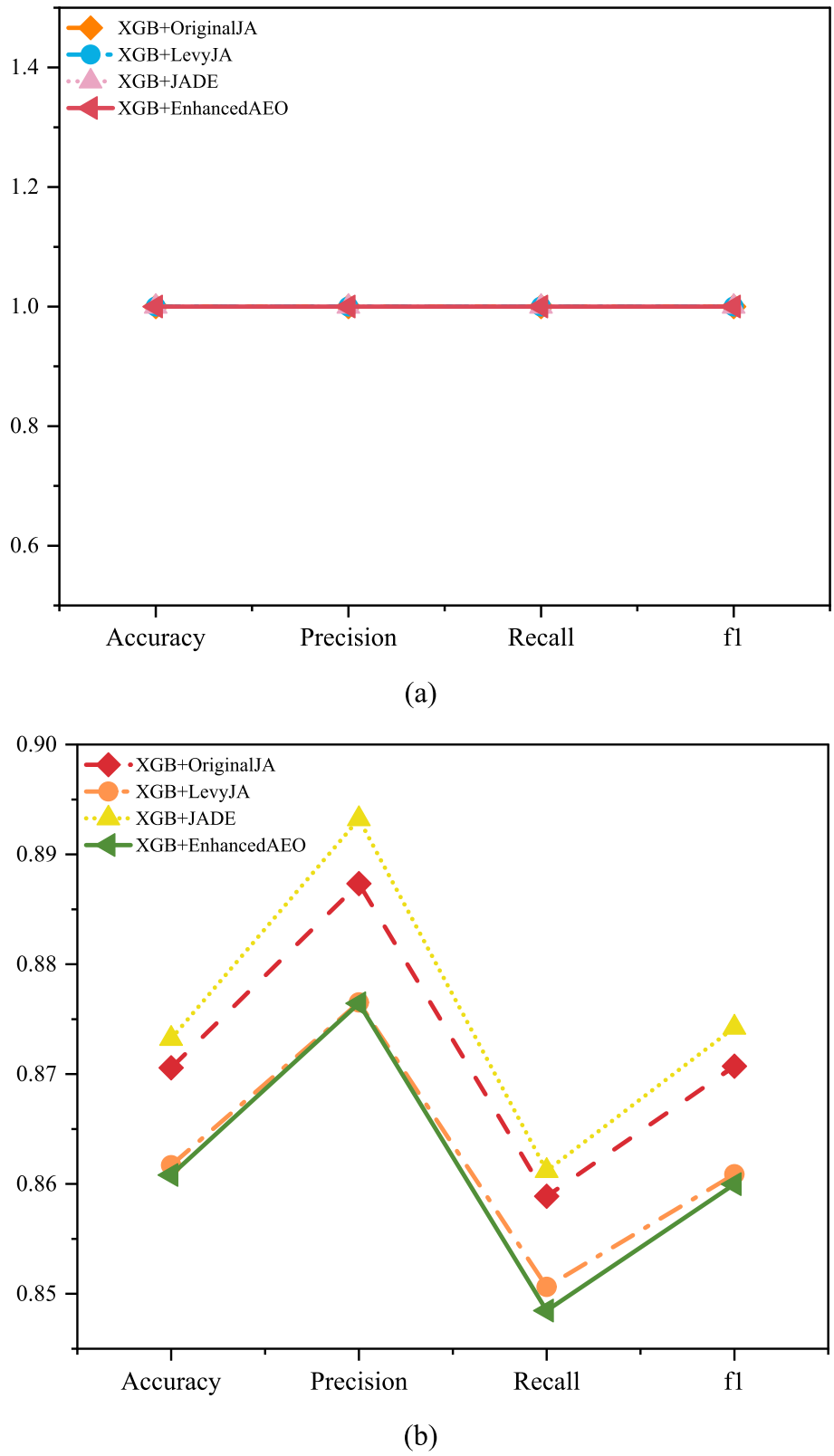

Fig. 6. The performance metrics obtained from XBboost, and XGboost with Original, LevyJA, JADE, and EnhancedAEO hybrid models from (a) training and (b) testing datasets.

performance metrics in this context. However, with the strategic integration of optimizer algorithms, a notable improvement in performance metrics becomes evident. Of particular significance is the augmented performance observed in the XGBoost model when paired with the JADE optimizer. This pairing consistently demonstrates superior performance across various performance metrics. Further analysis reveals that the JADE optimizer achieves optimal performance with specific parameter configurations. Notably, the identified optimal parameters for the JADE

optimizer are as follows: {' learning rate ': 0.3276316193519062, 'max depth': 8, ' N-estimators ': 225}. This finely tuned configuration underscores the efficacy of the JADE optimizer in enhancing the performance of the XGBoost classification model within the context of this study.

The comprehensive results for all epochs across various hybrid classifications of XGBoost paired with different optimizers were meticulously illustrated in Figs. 7 through 10. Upon analysis, it became evident that among the four hybrid models combining XGBoost with Original, LevyJA, JADE, and EnhancedAE, the standout performer was JADE. Remarkably, leveraging JADE led to an impressive F1 score of 0.8742, achieved after roughly 300 epochs. Notably, this high-performance level persisted consistently for an additional 200 epochs, showcasing the robustness and efficacy of the JADE optimizer in conjunction with XGBoost.

In light of the dataset's imbalanced nature, a comprehensive evaluation strategy was employed, leveraging macro-precision, macro-recall, and macro-F1 scores to gauge performance robustness. The visualization in Fig. 11 meticulously illustrates performance metrics, encompassing accuracy, precision, recall, and F1 score, for both default and optimized hyperparameters of the XGBoost model through metaheuristic optimization algorithms. Notably, precision emerges as the standout metric across all models, indicating a consistent strength in correctly identifying positive cases. Conversely, recall, while generally lower, signifies the model's ability to capture true positive instances among all actual positive cases. The optimization of XGBoost with JADE and OriginalJA hyperparameters notably elevates performance metrics, underlining the efficacy of metaheuristic techniques in fine-tuning model parameters for superior classification outcomes. Ultimately, the XGBoost model integrated with JADE emerges as the optimal choice for the ECG dataset, boasting an impressive F1 score of 0.8742. This significant enhancement, compared to Walsh's [1] findings with a macro F1 score of 0.82, underscores the substantial progress achieved in classification accuracy.

Moreover, macro metrics are employed through the optimization process, which neglects data imbalance and treats all classes in a uniform format. As a result, the sample number per class does not affect the averaging process. In addition, the initial objective of the optimization is to elevate the F1-score, which is calculated in a macro sense, which means that the sample count holds no significance. Consequently, the parameter optimization is directed towards the achievement of reliable prediction and classification through all classes. This approach contributes to the high accuracy observed in the outputs. More clarifications are carried out in Table 1. Based on the results, the accuracy is very high for each class, while the classes are imbalanced.

Although the main focus of this study is on improving accuracy in classification, it still remains highly motivational to address issues related to inference speed and computational efficiency with thoughts of its potential application in wearable health devices. Fast, low-latency processing is fundamentally required in real-time analytics of ECG in light of timely intervention in critical health scenarios. This is where big machine learning models, especially those from deep learning, face a bottleneck due to their computational complexity. Although XGBoost ranked very well among other machine learning methods in this study, it is still necessary to indicate how well they will work in practice. In fact, all such improvements in the inference speed of models may be achieved by techniques such as model pruning, quantization, and hardware acceleration using special processors like GPUs or TPUs. Further works should evaluate the model's performance in terms of both its accuracy and computational efficiency, to find a good compromise that can offer very high accuracy in the classification of electrocardiograms, with the further guarantee of model lightness and adequate speed for real-time applications on very resource-constrained wearable devices. Furthermore, the wearable device's computational requirement can also be reduced by exploring edge computing in addition to cloud processing, which enables the usage of even more sophisticated models with less latency. These can be further investigated for the practical implementation of the proposed system for continuous ECG monitoring in realistic healthcare environments.

Besides, while the present research has attained an appropriate level of accuracy for the PTB Diagnostic ECG Database, it should realize that generalization to a wide variety of data sets indeed guarantees robustness and applicability in reality. The diversity of results in different

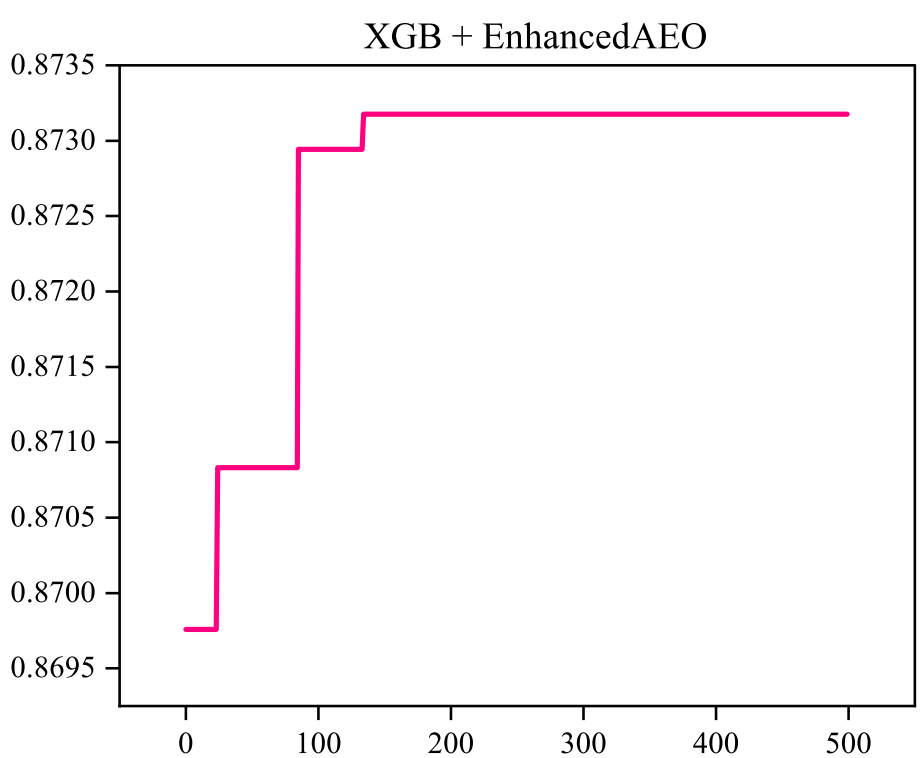

Fig. 7. The results of all epochs of the XGboost and EnhancedAEO classification model.

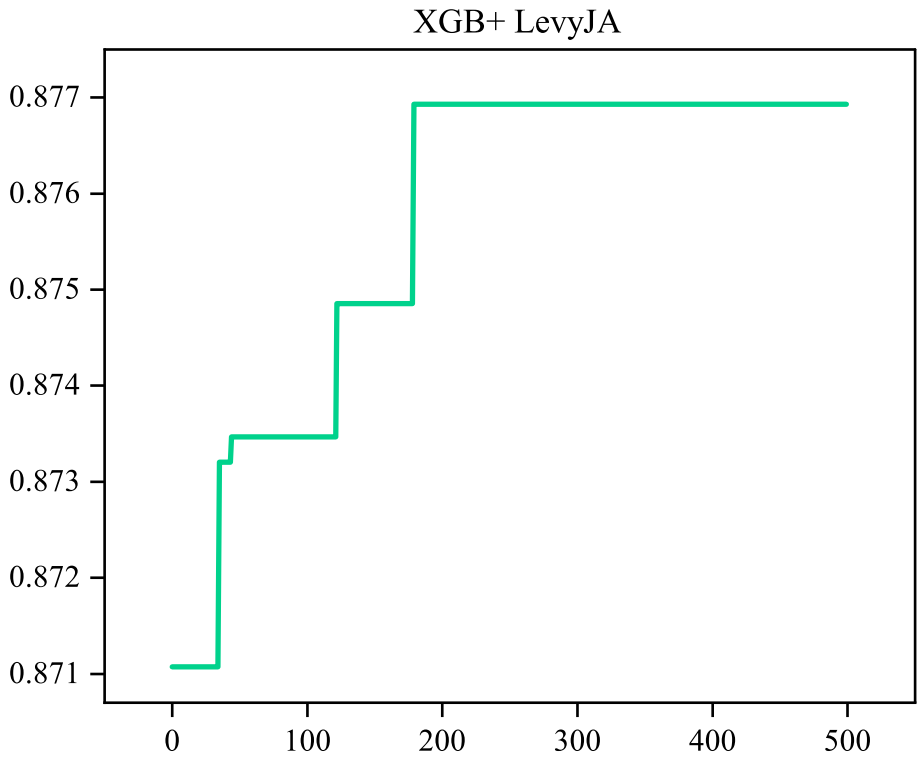

Fig. 8. The results of all epochs of the XGboost and LevyJA classification model.

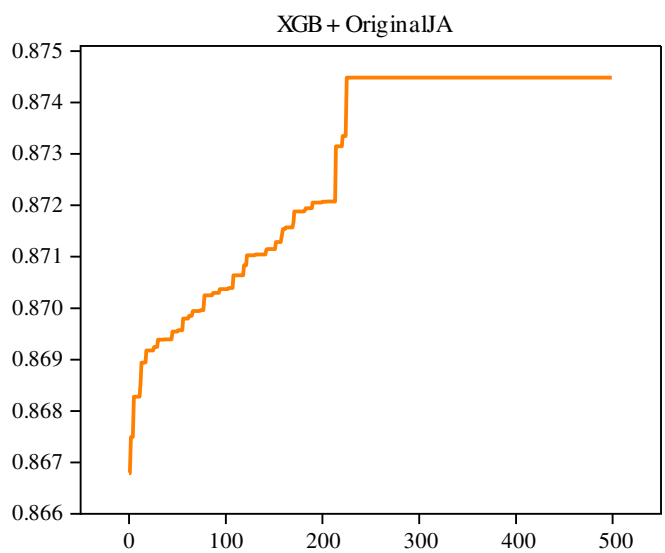

Fig. 9. The results of all epochs of the XGboost and OriginalJA classification model.

performances among multiple machine learning models may be performed within data coming out of datasets that have different properties, such as modifications to signal quality, patient demographics, and the emergence of different kinds of arrhythmias. A deeper analysis covering numbers of ECG datasets, such as the MIT-BIH Arrhythmia Database, INCART Database, and other similar datasets, will definitely yield a more reliable evaluation of generalization performance. This study can also be expanded to discuss the performance of the proposed model on these various datasets and the possible use of transfer learning or domain adaptation for better generalization performance in datasets. It is also very important that this model can assure accuracy and efficiency across a variety of populations and signal characteristics, and practical deployments across global healthcare systems. The future study will also improve by expanding the pool of data to improve robustness and effectiveness in a clinical setting rather than a controlled experimental environment.

4. Conclusion

In this research endeavor, we meticulously conducted the classification of ECG datasets utilizing four distinct models: Support Vector Classifier (SVC), RandomForest, XGBoost, and LinearSVC. Prior to deploying these models, Principal Component Analysis (PCA) was adeptly employed to streamline the dataset's complexity, reducing it to 30 features. This preprocessing step played a pivotal role in enhancing the models' performance by effectively managing data intricacies. The evaluation of model performance was anchored on the F1 score, a metric deemed more reliable for unbalanced datasets compared to accuracy. This comprehensive metric considers both precision and recall, providing a more holistic assessment of model performance. Results indicated XGBoost as the top-performing model among the four, showcasing its superiority in accurately classifying ECG datasets. To further bolster the XGBoost model's performance, advanced metaheuristic optimization algorithms – including Enhanced AEO, LevyJA, JADE, and OriginalJA – were judiciously employed to fine-tune critical hyperparameters. This meticulous optimization process aimed to elevate the XGBoost model's classification prowess beyond its default configuration.

In subsequent analyses, detailed evaluations were conducted to scrutinize the impact of PCA preprocessing, model performance, and hyperparameter optimization on the XGBoost model's classification efficacy. Notably, the results offered comprehensive performance metrics post PCA analysis, revealing XGBoost's standout performance among the models assessed.

Further, the integration of metaheuristic optimization algorithms yielded discernible improvements in the XGBoost model's classification performance. Notably, the JADE optimizer emerged as the standout performer, consistently enhancing the XGBoost model's efficacy across

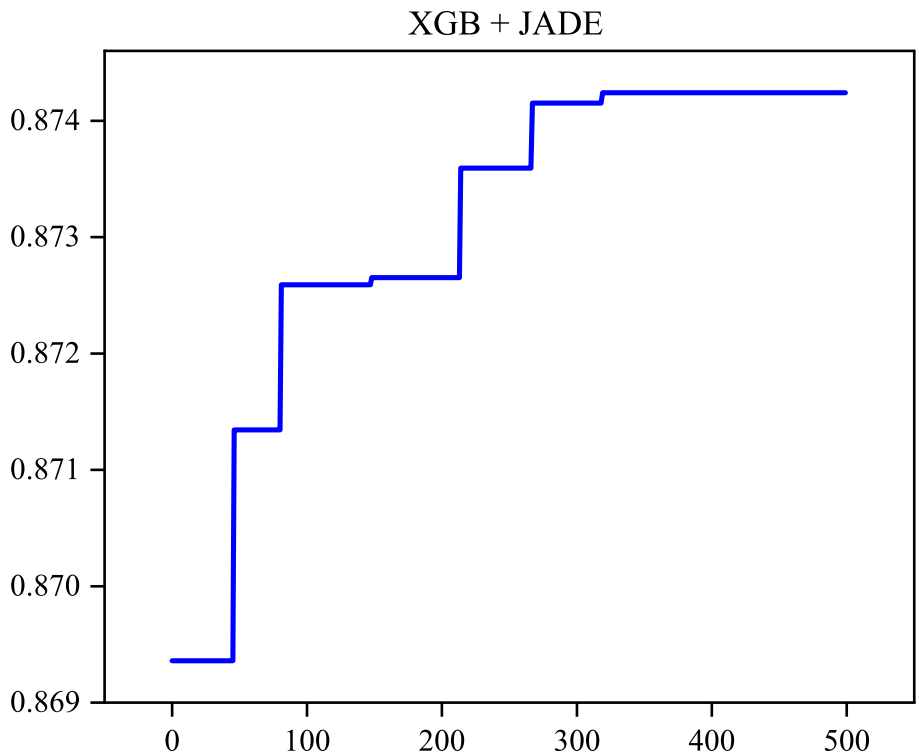

Fig. 10. The results of all epochs of the XGboost and JADE classification model.

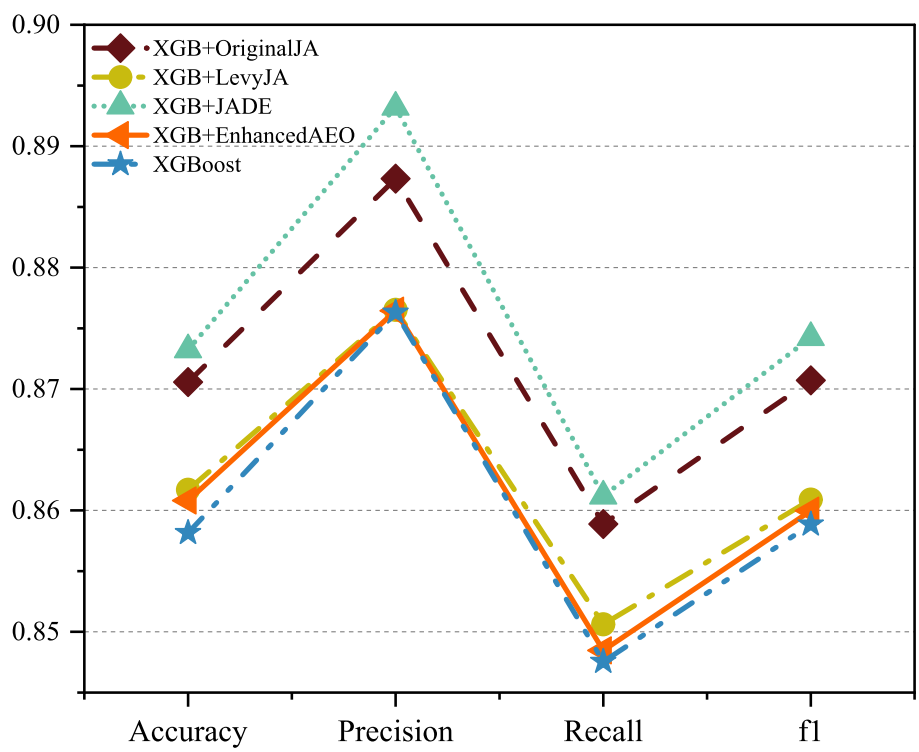

Fig. 11. The performance metrics obtained from XBboost, and XGboost with Original, LevyJA, JADE, and EnhancedAEO hybrid models.

various performance metrics.

A meticulous examination of results across different epochs underscored the robustness of the XGBoost model when paired with the JADE optimizer, achieving an impressive F1 score of 0.8742. This notable performance enhancement, compared to prior findings, highlights the substantial progress achieved in classification accuracy. Ultimately, the XGBoost model augmented with JADE emerges as the optimal choice for ECG dataset classification, showcasing significant advancements in classification accuracy and model robustness.

Table 1
The values of metrics through various classes, macro, and weighted average approaches.

Approach	Precision	Recall	F1-Score	Accuracy	Support
Class 0.0	0.99	0.97	0.98	0.86	18,118
Class 1.0	0.80	0.76	0.78	0.86	556
Class 2.0	0.81	0.95	0.87	0.86	1448
Class 3.0	0.82	0.74	0.78	0.86	162
Class 4.0	0.82	0.95	0.88	0.86	1608
Macro Avg.	0.89	0.86	0.87	0.86	21,892
Weighted Avg.	0.96	0.96	0.96	0.86	21,892

CRediT authorship contribution statement

Keyi Tang: Formal analysis, Data curation. **Shuyuan Ma:** Methodology, Funding acquisition. **Xiaohui Sun:** Validation, Software, Methodology. **Dongfang Guo:** Writing – original draft, Supervision, Project administration, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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